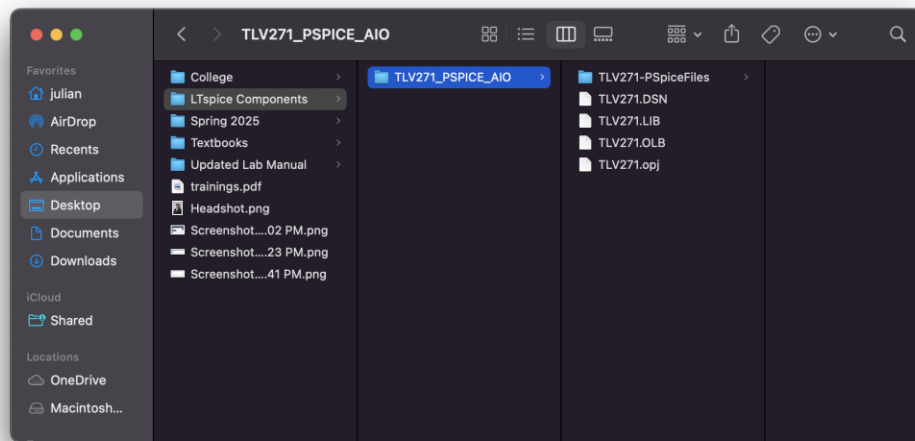


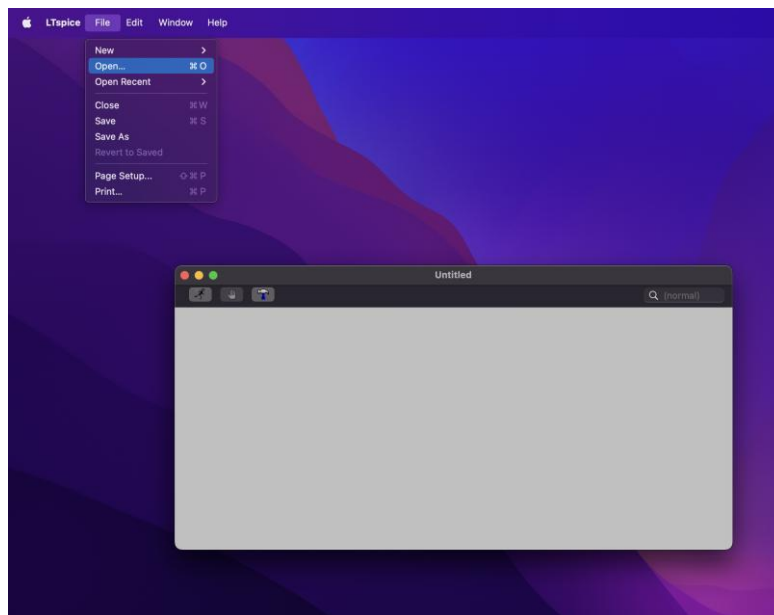
EEL 3111C How to import components in LTspice: Mac Edition

If you are struggling to follow along with the video instructions from Lab 5 or 8, this document should hopefully help.

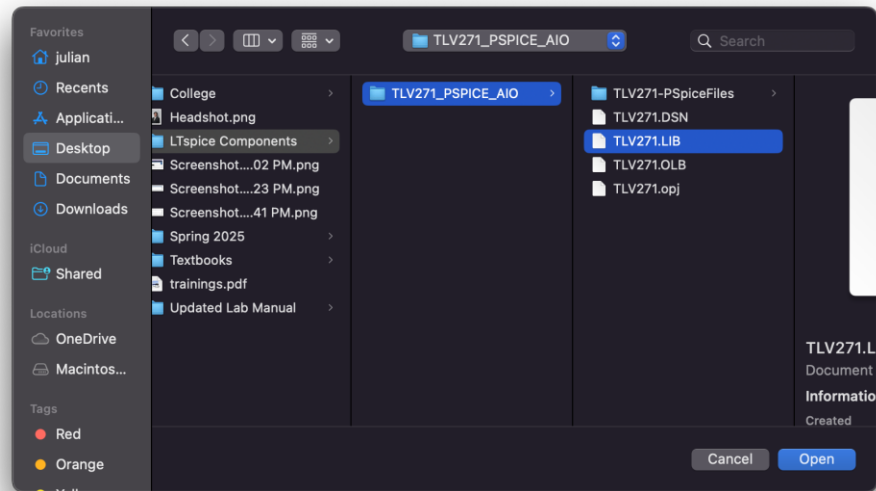
1. Download the zip file from canvas. For the op amp in lab 5: "slom249.zip". For the comparator in lab 8: "LM393sljc016.zip". These zip files should automatically be saved to your Downloads folder. Extract the contents of the zip file by right clicking, then save these files to a directory you have access to. For simplicity, you can create a folder on your desktop and save all component files there. The image below is for the op amp from lab 6.



2. Open up LTspice and open a new schematic. From there you can click File->Open or press Command+O.



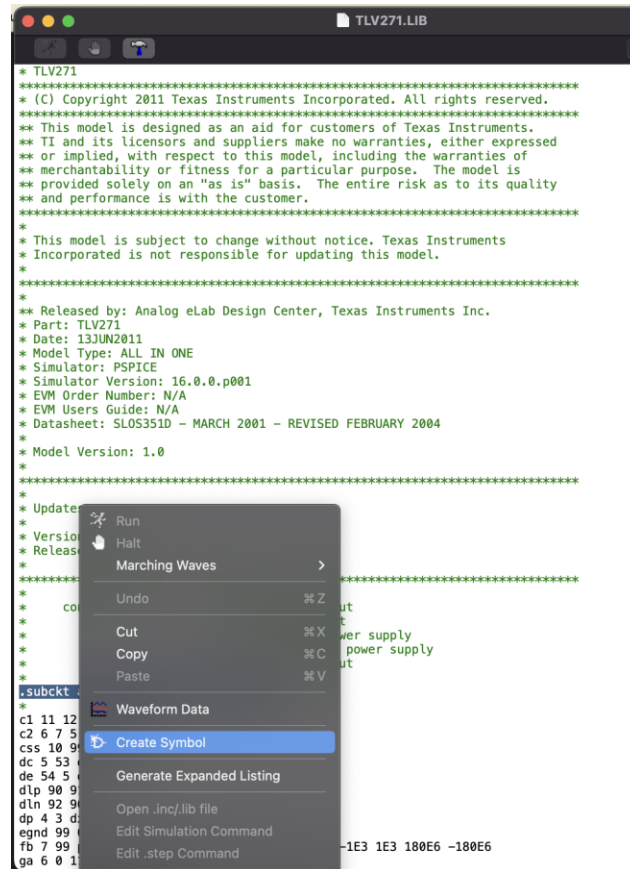
3. Navigate to the extracted file, and look for the “.LIB” file. For the comparator in lab 8, you open the LM393.5_1 file.



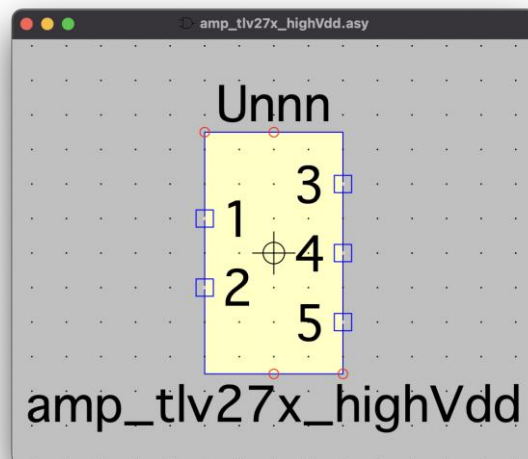
4. Once you open the file, you should see a window pop up. This tells you all the info you need about the component, including the connections, model parameters, and datasheet name if needed.

```
TLV271.LIB
* TLV271
* (C) Copyright 2011 Texas Instruments Incorporated. All rights reserved.
*****
** This model is designed as an aid for customers of Texas Instruments.
** TI and its licensors and suppliers make no warranties, either expressed
** or implied, with respect to this model, including the warranties of
** merchantability or fitness for a particular purpose. The model is
** provided solely on an "as is" basis. The entire risk as to its quality
** and performance is with the customer.
*****
* This model is subject to change without notice. Texas Instruments
* Incorporated is not responsible for updating this model.
*
*****
** Released by: Analog eLab Design Center, Texas Instruments Inc.
** Part: TLV271
** Date: 13JUN2011
** Model Type: ALL IN ONE
** Simulator: PSPICE
** Simulator Version: 16.0.0.p001
** EVM Order Number: N/A
** EVM Users Guide: N/A
** Datasheet: SL05351D - MARCH 2001 - REVISED FEBRUARY 2004
*
* Model Version: 1.0
*
*****
* Updates:
*
* Version 1.0 :
* Release to Web
*
*****
* connections:      non-inverting input
*                   | inverting input
*                   | | positive power supply
*                   | | negative power supply
*                   | | output
*                   | |
* .subcckt amp_tlv27x_highVdd 1 2 3 4 5
*
c1 11 12 457.48E-15
c2 6 7 5.0000E-12
css 10 99 1.1431E-12
dc 5 53 dy
de 54 5 dy
d1p 90 91 dx
d1n 92 90 dx
dp 4 3 dx
egnd 99 0 poly(2) (3,0) (4,0) 0 .5 .5
fb 7 99 poly(5) vb vc ve vlp vln 0 176.02E6 -1E3 1E3 180E6 -180E6
ga 6 0 11 12 16.272E-6
```

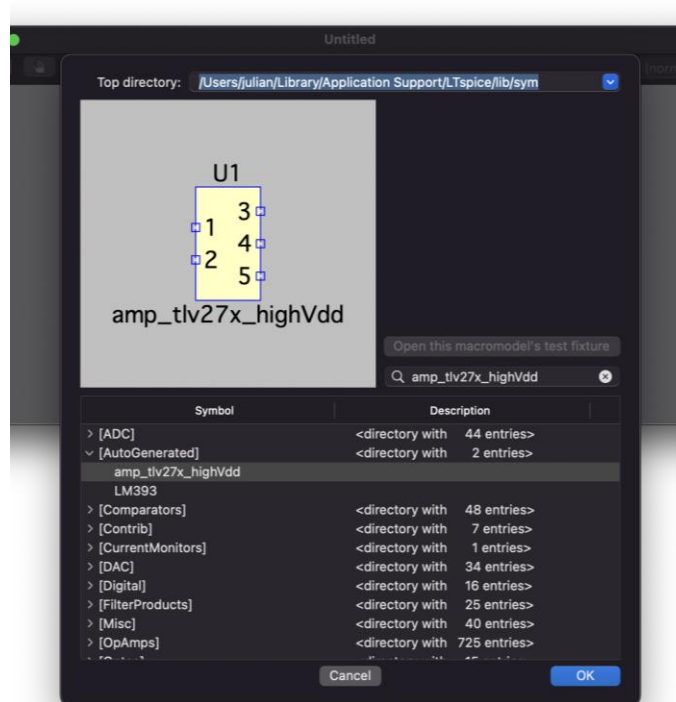
- To create the new symbol, highlight the blue text “.subckt amp_tlv27x_highVdd 1 2 3 4 5” and right click to press Create Symbol.



- The following should now appear on your screen. That’s your new symbol! If you wish to change the shape, just google tutorials online but it might cause you errors. In case you mess up and can’t figure out what’s wrong, just delete the files and restart again and use the default shape.



- Go back to your schematic and start building your circuit. To access your new component, right click the screen->Draft->Component and then open the AutoGenerated folder. Your component should be there.



- Press ok and place your component down. Continue to build your circuit as normal. You did it! Below is an example of an inverting op amp circuit. For the comparator, all steps are the same after you open the "LM393.5_1" file.

